2020: 1(c)

Given that,

input values	desired output values
$x^1 = (1\ 0\ 1)^T$	-1
$x^2 = (0\ -1\ -1)^T$	1
$x^3 = (-1\ -0.5\ -1)^T$	1

where, weight vector, $w_0 = (1\ -1\ 0)^T$

learning rate $= 0.1$

The perceptron learning is actually intended to find the correct weight values by training with the training set. After feeding each input, the weight will become more accurate. The weight update can be done by using the relation:

$$w_{i+1} = w_i + c\,(\text{desired output} - \text{actual output})\,x$$

(+) (a)

- For $x^1(1, 0, 1)^T$:

actual output, $a = w_x x_1 + w_y x_0 + w_2 x_1$

$= 1 \times 1 + (-1) \times 0 + 0 \times 1$

$= 1 > 0 \quad (\text{hardlims}(1) = 1)$

$= 1$

desired output, $t = -1$

Now the weights will be updated as,

$w_x = 1 + 0.1(-1-1) \times 1 = 0.8$

$w_y = -1 + 0.1(-1-1) \times 0 = -1$

$w_2 = 0 + 0.1(-1-1) \times 1 = -0.2$

$w = [0.8 \quad -1 \quad -0.2]$

- For $x^2(0, -1, -1)^T$

$a = 0.8 \times 0 + (-1) \times (-1) + (-0.2) \times (-1) = 1.2 > 0 = 1$

$t = 1$

so, weight will remain same as before i.e. no change

$w = [0.8 \quad -1 \quad -0.2]^T$

- For $x^3 (-1, -0.5, -1)^T$

hardlims(-0.1) = -1

$a = 0.8 \times -1 + (-1) \times (-0.5) + (-1) \times (-0.2) = -0.8 + 0.7 = -0.1 < 0 = -1$

$t = 1$

updated weight:

$w_x = 0.8 + 0.1 (1 - (-1)) (-1) = 0.6$

$w_y = -1 + 0.1 (1+1) (-0.5) = -1.1$

$w_z = -0.2 + 0.1 (1+1) (-1) = -0.2 - 0.2 = -0.4$

so, $w = [0.6 \quad -1.1 \quad -0.4]$

This procedure will continue till it will be in steady state.

Next iteration:

For $x^1 (0, 1, 1)^T$

$a = 0.6 \times 1 + (-1.1) \times 0 + (-0.4) \times 1 = 0.2 > 0 = 1$

$t = -1$

so, $w = [0.4 \quad -1.1 \quad -0.6]^T$

- For $x^2(0, -1, -1)^T$

$a = 0.4 \times 0 + (-1.1) \times (-1) + (-0.6) \times (-1) = 1.7 > 0 = 1$

$t = 1$

(no change) $\rightarrow w = \begin{bmatrix} 0.4 & -1.1 & -0.6 \end{bmatrix}$

- For $x^3(-1, -0.5, -1)^T$

$a = 0.4 \times (-1) + (-1.1) \times (-0.5) + (-0.6 \times (-1) = 1$

$t = 1$

... no change

- again check for $x^1(1, 0, 1)^T$

$a = 0.4 \times 1 + (-1.1) \times 0 + (-0.6) \times 1$

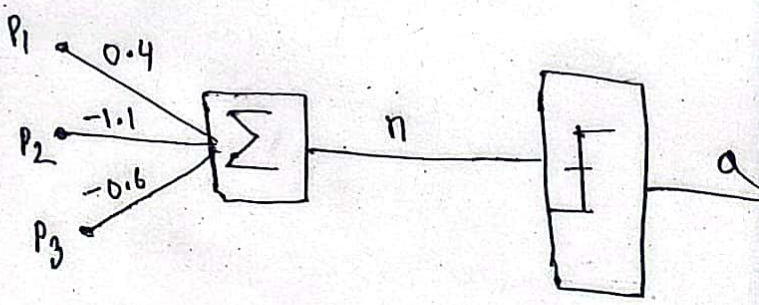
$= 0.4 - 0.6 = -0.2 < 0 = -1$

$t = -1$, no change,

so, the final weight:

$w = \begin{bmatrix} 0.4 & -1.1 & -0.6 \end{bmatrix}^T$

Neural network:



$a = hardlim(wp)$